host_agent

The largest lake in Exeter has an area of approximately 21,861.73 m² and is located at about (50.728481, -3.546139). It is classified as "Still Water." A map has been generated showing the location and details of this lake.

planning_agent

The area specified is Exeter, so there will be 1 entry for Exeter. The entity to search for is lakes, and the condition is to find the largest lake specifically.

output steps: ["Find Exeter 1 area", "Search for lakes in Exeter as many entries as possible", "Identify the largest lake from the search results"]

named_area_agent

The specific area polygon for Exeter has been filtered and saved as 'filtered_exeter_area_polygon'.

data_analysis_agent

Filtered Exeter area polygon(s): 1 entries found. An artifact has been generated with the name 'filtered_exeter_area_polygon' and the description: Polygon(s) from exeter_area_polygon corresponding to Exeter city or settlement.

Tool Code Executor

```
def filter_exeter_area_polygon(data: list):
    gdf = data[0]
    # Filter for Exeter using multiple relevant columns
    # Use name1_text for 'Exeter', and also check for descriptiongroup and description for settlement/city
    filtered = gdf[((gdf['name1_text'].str.lower() == 'exeter') |
                    (gdf['name1_text'].str.lower().str.contains('exeter'))) &
                   (gdf['descriptiongroup'].isin(['Settlement', 'Part Of Settlement'])) &
                   (gdf['description'].isin(['City', 'Town', 'Suburban Area']))]
    # If not found, try a broader filter for any polygon with 'exeter' in name1_text
    if filtered.empty:
        filtered = gdf[gdf['name1_text'].str.lower().str.contains('exeter', na=False)]
    summary = f"Filtered Exeter area polygon(s): {len(filtered)} entries found."
    artifact_name = 'filtered_exeter_area_polygon'
    artifact_description = 'Polygon(s) from exeter_area_polygon corresponding to Exeter city or settlement.'
    return [summary, artifact_name, artifact_description, filtered]
```

water_features_agent

I found 70 records of lakes within the 'filtered_exeter_area_polygon'. The artifact generated is:

- **Artifact Name:** `waterpoint_lakes_in_exeter`
- **Description:** A geopandas dataframe containing water point data with bbox applied for the query with filters applied.

If you need further analysis, please let me know!

plotting_agent

The largest lake in Exeter has an area of 21,861.73 m² and is located at approximately (50.728481, -3.546139). Here are some details of the lake: osid: b59c379f-a8ac-4e52-87a7-45d877766295, toid: osgb1000000347764329, description: Still Water, oslandcovertiera: Water.

A folium map has been generated showing the largest lake, its location, and area.

- Artifact name: largest_lake_map
- Artifact description: Folium map showing the largest lake in Exeter, with its location and area.
- Map file: largest_lake_map.html

Tool Code Executor

```
def plot_largest_lake(data: list):
    import geopandas as gpd
    import folium
    from shapely.geometry import Polygon

    gdf = data[0].copy()
    # Filter for lakes: description should be 'Still Water', oslandcovertiera 'Water', oslandcovertierb 'Inland Water', watertype 'Inland'
    lakes = gdf[(gdf['description'] == 'Still Water') &
                (gdf['oslandcovertiera'] == 'Water') &
                (gdf['oslandcovertierb'] == 'Inland Water') &
                (gdf['watertype'] == 'Inland')]

    # Find the largest lake by area
    largest_lake = lakes.loc[lakes['geometry_area_m2'].idxmax()]
    largest_lake_gdf = gpd.GeoDataFrame([largest_lake], crs=lakes.crs)
    largest_lake_gdf = largest_lake_gdf.to_crs(epsg=4326)
```

```
    # Get centroid for marker
    centroid = largest_lake_gdf.geometry.iloc[0].centroid

    # Create folium map
    m = folium.Map(location=[centroid.y, centroid.x], zoom_start=15)
    folium.GeoJson(largest_lake_gdf.geometry.iloc[0], name='Largest Lake').add_to(m)
    folium.Marker([centroid.y, centroid.x],
                  popup=f"Largest Lake\nArea: {largest_lake['geometry_area_m2']:.2f} m²").add_to(m)

    map_filename = 'largest_lake_map.html'
    m.save(map_filename)

    summary = f"The largest lake has area {largest_lake['geometry_area_m2']:.2f} m². Location: ({centroid.y:.6f}, {centroid.x:.6f}). First 5 columns: {largest_
    artifact_name = 'largest_lake_map'
    artifact_description = 'Folium map showing the largest lake in Exeter, with its location and area.'
    return [summary, artifact_name, artifact_description, map_filename]
```